

Landscape Particular Specification

FOR

PROPOSED DATA CENTRE DEVELOPMENT

**AT NO. 2-10 TAI YUEN STREET,
KWAI CHUNG, N.T.
ON KWAI CHUNG TOWN
LOT NO. 161**

**PREPARED BY
EARTHASIA LIMITED**

APR 2022

Landscape Particular Specification

SECTION - I

A. General

B. Demolition, Site Clearance & Alterations

SECTION – II

A. Soft Landscape Works

- I. DEFINITION AND SPECIFICATION OF MATERIALS**
- II. PREPARATORY WORKS**
- III. PLANTING WORKS: GENERAL**
- IV. PLANTING TECHNIQUES: TREES**
- V. EXISTING TREES**
- VI. ESTABLISHMENT WORKS**

B. Living Green Wall

SECTION - I

A. General

GENERAL

1.0 GENERAL SPECIFICATION AND PARTICULAR SPECIFICATION

- (1) This Particular Specification shall be read in conjunction with the General Specification for Building by Architectural Services Department (2017 Edition) and all current amendment thereto (herein referred to as 'the G.S.').
- (2) All clauses of the G.S. shall be applicable to this Contract unless expressly stated in this Particular Specification. Provided where there are any discrepancies between the G.S. and this Particular Specification, the latter or other the higher standards shall take precedence.
- (3) This Particular Specification shall be applied to landscape works.

2.0 EQUIVALENT PRODUCT

- (1) The Contractor may use the proprietary products mentioned in the Drawings and/or Specifications or other products having proven equivalent functions or performance, subject to the approval by the Architect and the Landscape Architect.
- (2) If the Contractor intends to use the intellectual property rights of another party in performing his obligations under the Contract, appropriate licenses should be obtained from the relevant owners.

3.0 REFERENCE WITH ARCHITECTURAL WORKS

- (1) This Particular Specification shall be read in conjunction with Particular Specifications for Architectural Works and other relevant disciplines where applicable.

B. Demolition, Site Clearance & Alterations

Demolition, Site Clearance & Alterations

1.0 GENERAL REQUIREMENTS

- (1) Contractor shall take note of all specifications and requirements pertaining to demolition, site clearance and alterations works mentioned in other parts of the contract, including but not limited to architecture, engineers, environmental consultant's drawings and specifications.
- (2) The Contractor shall comply with all requirements on tree preservations, protection and tree felling methodology in:
- (3) Particular specification for Tree Transplanting and Felling to ensure demolition, site clearance and alterations works will not damage preserved trees in the site and surroundings. Monitoring and regular tree maintenance shall be carried out to the satisfaction of the site staff or the Landscape Architect.

2.0 SUBMISSIONS

- (1) The Contractor shall report to the Architect immediately for any discrepancy between Tree Survey Plans, Existing Tree Schedule, and existing tree on site
- (2) Prior commencement of site works the Contractor shall follow all of the following but not limited to programming, provide site supervisory staff and preservation and protection of existing trees, provide tree survey record, and provide photographic record of existing trees.
- (3) Please refer to details of the submission in:
 - a. Particular specification for Tree Transplanting and Felling – Section 2
 - b. Existing Vegetation Treatment Including Protection and Surgery

3.0 WORKMANSHIP

- (1) The Contractor shall commence demolition site clearance and alteration works only after the Contractor has complied with requirements to complete works prior to commencing site clearance, demolition, construction of any permanent or temporary works, or any other site operations. The Contractor is required to meet the following and the requirements under Particular specification for Tree Transplanting and Felling – Section 1-5.
- (2) General Requirements are as follows:
 - a. Submission of survey plans of existing trees, record photographs, and any discrepancies and have obtained reply to the discrepancies and approval of the survey plans and record photographs.

4.0 INSTALLATION OF TREE IDENTIFICATION LABELLING

- (1) Agreement with the site staff and/or Landscape Architect the extend of clearance of existing trees
- (2) Temporary protective fencing around trees that will be retained shall be constructed.
- (3) Temporary application of protective layer such as mulching within the temporary protective fencing area around the retained tree
- (4) For all detailed requirements, please refer to Particular specification for Tree Transplanting and Felling – Section I-V.

SECTION – II

A.Soft Landscape Works

SCOPE OF TREE PLANTING WORKS

The Tree Planting Works to be carried out by the Contractor under this Particular Specification comprise:

A. Tree Planting Works:

The scope comprised planting of trees and preparation as shown or referred to this Particular Specification and Planting Schedule (refer to **Drawing LN-3.01**). The location of tree planting refers to landscape drawings, the Contractor is responsible for supply, prepare and install all materials specified in this Particular Specification and coordinate with the Architect and the Landscape Architect to commence specified tree planting works.

B. Removal of Existing Tree Stub if Found Any

The scope comprised removing all existing tree stub found at specified planting locations, the extent of the removal works is measured by 5 meters radius from the planting locations specified on drawings. The Contractor is responsible for providing all necessary labour and machinery to carry out the existing tree stub removal works and dumping.

C. Submission:

The scope comprised submitting method statements for preparation works, tree protection, tree planting and maintenance works, planting materials samples and all necessary test reports.

The submission of planting material shall be in color printed A4 paper size format, indicating the Latin name and Chinese name of the planting material with photos clearly showing the measurement of height, spread and diameter at breast height. The approval of the material submission is given to site delivery condition and onsite inspection by the Landscape Architect. Whenever the delivered material(s) is deviated from the material submission or regarded as poor quality in the Landscape Architect's opinion, the Contractor is required to resubmit the material(s) and remove delivered material(s) at his own cost.

All submissions shall be 30 days within the award of contract to the Landscape

Architect for approval; no planting works shall be carried out unless the approval has been given to the Contractor or directly instructed by the Landscape Architect.

APPROVED LIST OF SOFT LANDSCAPE CONTRACTOR

1. Asia Landscaping Limited
2. Oriental Landscapes Limited
3. Tarzan Landscape Contractors Limited
4. Tsui Yuen Garden Limited

I. DEFINITIONS AND SPECIFICATION OF MATERIALS

1.0 DEFINITIONS

(a) Soils

- (i) Subsoil is friable, fine grained, decomposed granite or volcanic containing no stones larger than 50 mm diameter.
- (ii) Topsoil shall be as defined in Clause 1.1 in this section.
- (iii) Topsoil Mix is an artificially constituted topsoil.
- (iv) For the purposes of this Section Completely Decomposed Granite (C.D.G.) is granite which has been completely decomposed by weathering in place, the texture of which is still recognizable.

(b) Cultivation

- (i) Cultivation is the controlled decompaction of the upper layers of soil in order to provide an even textured friable growing medium sufficiently open to allow air penetration and water retention so as to provide favorable conditions for plant growth.
- (ii) Ripping is the controlled decompaction of the upper layers of soil by a tractor drawn tine or plough.
- (iii) A Fine Tilth is a friable soil as a result of cultivation.
- (iv) Scarifying is the loosening of soil using a pronged implement without turning of soil.

(c) Soil Conditioner is an organic mixture added to mineral soil to improve existing soil structure and texture.

(d) Mulch is a stable fibrous organic material spread as a surface treatment to reduce soil erosion and water loss.

(e) Fertilizer is an organic and chemical mixture containing nitrogen (N). Phosphorous (P), potassium (K) and other specified elements in stated quantities

to promote plant growth.

Soil Binding Agent is a material which stabilizes and conditions soil and aids moisture retention.

A Container is a rigid or semi-rigid receptacle, with drainage holes at the base, in which plants are grown.

- (h) A Containerized Plant is a plant which has been grown in a container for not less than three months in the growing season before planting out and which is delivered in that container for planting out.
- (i) An Open Ground Plant is a plant which has been grown by planting out into natural ground.
- (j) The Rootball of a plant is the firmly bound fibrous root and soil which is removed intact with the plant.
- (k) The Root Collar is the position on the plant stem where the plant naturally from the ground when growing in the nursery.
- (l) Plant Descriptions
- (n) Undercutting is the operation of serving downward growing plant roots in-situ to control root development.
- (o) Root Pruning is the cutting away of part of the root system to stimulate fibrous root growth within a compact ball.
- (p) Protective Material is a covering mesh that provides protection to the treated surface.
- (q) A Berm is a flat section of a terrace or the continuous artificial contouring of a slope.
- (r) Erosion is the detachment and movement of the soil or rock fragments by natural forces.

1.1 SOILS

- (a) Imported Subsoil shall be free from all impurities including chemicals, oils, cement, sticky clays and stones exceeding 50mm in diameter. Subsoil shall be submitted to the Architect or Architect's Representative for approval before importation and use.
- (b) Imported topsoil shall be evenly texture, fertile, dark brown or black colour, medium loam. The soil shall be free from weeds, deleterious matter and stones larger than 50mm in any dimension. It shall not be excessively sandy, gritty or water logged.

The imported topsoil shall be analysis for N.P.K. value, organic matter content, Cation Exchange Capacity ratio, organic carbon, pH value, physical content of sand, silt and clay, and water content. Soil testing shall be arranged by the Contractor and carried out by an approved reputable firm or institute at the contractor's cost, and the report shall be submitted to the Architect or Architect's Representative for approval.

The imported topsoil shall comply with the following:

pH value	:	5.5 - 7.5
% Deacetylation	:	60 – 70%
% organic matter	:	not less than 7.5%
% organic carbon	:	2.0 - 3.0
% total nitrogen	:	0.09 - 0.15
Carbon / Nitrate ratio	:	25:1 - 45:1
Available P 0	:	7mg/100g - 10mg/100g
Exchangeable K 0	:	15mg/100g - 30mg/100g
Cation Exchange Capacity	:	16 -20 m.e. %
Viable Count	:	not less than 3000/g

Where the imported topsoil does not meet the above criteria it may be modified by the addition of peat, fertilizer and lime in quantities to be approved by the Architect or Architect's Representative.

The Contractor shall supply a representative 1kg sample of approved topsoil to the Architect or Architect's Representative. Approval to the sample must be

obtained before bulk delivery commences, and approval of the sample will not preclude the right of the Architect or Architect's Representative to reject any imported topsoil which in the opinion of the Architect or Architect's Representative falls appreciably below the standard of the sample. The sample shall be retained by the Architect's Representative in a location on Site which will allow inspection and comparison throughout the period of the Contract.

No Change in the source of imported topsoil shall be allowed without the prior approval of the Architect or Architect's Representative based on such tests and samples as specified herein.

- (c) The Contractor shall inform the Architect or Architect's Representative if he is unable to supply topsoil which complies with the specification. In this case substandard topsoil or completely decomposed granite (C.D.G.) to which soil conditioner has been added may be approved as a substitute, forming a "Topsoil Mix" and "Light Weight Soil". "Topsoil Mix" and "Light Weight Soil" shall comprise substandard topsoil or completely decomposed granite (C.D.G.) to which soil conditioner has been added. Soil conditioner shall be mixed with any substandard topsoil or C.D.G. at the following rates :

"Topsoil Mix"

Topsoil to comprise a mixture of :-

- i. 1 part by volume of manufactured soil conditioner
- ii. 1 part by volume of C.D.G or sub-standard topsoil
- iii. 0.5kg/M3 of water – retaining crystals

"Light Weight Soil Mix" (Imply the kpa/m3 in terms of light weight compare with topsoil)

Light weight soil to comprise a mixture of :-

- i. 1 part by volume of manufactured soil conditioner
- ii. 1 part by volume of C.D.G or sub-standard topsoil
- iii. 1 part by volume of vermiculiter or volcanic pellets, with a particle size of 5-10mm
- iv. 0.2kg/M3 of water – retaining crystal
- v. Total weight of the Light Weight Soil Mixture shall be 1000-1200kg/m3

The Contractor shall prepare topsoil mix on site. Mixing shall not take place during periods of heavy rain, nor when the soil is saturated. Mixing operations shall cease if the moisture content of the topsoil is too high to achieve thoroughly mixing of the conditioner with mixsoil.

Topsoil mix & Light Weight Soil shall be free from grass or weed growth of any kind, sticky clays, stones larger than 50mm in diameter, or other foreign matter. A sample of one cubic metre shall be submitted to the Architect or Architect's Representative for approval before importation and use.

Water Retaining Crystals or products having equivalent functions or performance as detailed as below, shall be applied when prepared topsoil mix:

- i. Chemical Type: Synthetic, high molecular weight, X-linked polyacrylamide
 - ii. Appearance: White, free-flowing, angular granules, 80% in the size range 500-2800 μ . Drybulk density: 0.76 approx.
 - iii. Moisture Content: 5-7% w/w as supplied.
 - iv. Solubility: Insoluble in water, alcohol, acetone, ether and most other organic solvents.
 - v. Absorbency: Absorbs water of wide-ranging chemical composition to form a stable gel of discrete particles.
 - vi. Water holding capacity in litres/Kg dry product
 - vii. De-ionised Water: 400
 - viii. 10% Sea Water: 50
- (d) Existing Topsoil shall be derived from the site strip. It shall be a free draining material of a sandy loam character and shall contain no grass or weed growth of any kind or other foreign material. It shall be evenly texture, fertile, dark brown or black in colour. It shall not be excessively sandy, gritty or waterlogged.

The Contractor in tendering shall deemed and to have inspected the existing topsoil and satisfied himself that it is acceptable for the work specified herein. No claims shall subsequently be considered in respect of this material which would relieve the Contractor of his responsibilities.

The Contractor should submit to the Architect a chemical and mechanical analysis of the existing soils as outlines below:

- i. pH value
- ii. % organic matter
- iii. % organic carbon
- iv. % total nitrogen
- v. Carbon / nitrate ratio
- vi. Available P2O5
- vii. Exchangeable K2O
- viii. Cation exchange capacity
- ix. % sand
- x. % silt
- xi. % clay
- xii. % water content

The topsoil shall comply with the following :

- i. pH value : 5.5 - 7.5
- ii. % organic matter : not less than 7.5%
- iii. % organic carbon : 2.0 - 3.0
- iv. % total nitrogen : 0.09 - 0.15
- v. Carbon / nitrate ratio : 25:1 - 45:1
- vi. Available P2O5 : 7 mg/100g 10mg/100g
- vii. Exchangeable K2O : 15mg/100g - 30mg /100g
- viii. Cation exchange capacity : 16 - 20m.e.%

Where the topsoil does not meet the above criteria it may be modified by the addition of peat, fertilizer and lime in quantities to be approved by the Architect or Architect's Representative.

The Contractor shall give the Architect four week notice of his intention to commence deposition of top soiling operations in order to allow for the results of the analysis to be available before commencing top soiling.

If the period between the analysis of the existing soils as above and the commencement of any deposition of topsoil exceeds 12 months then the Contractor shall carry out a second analysis of the topsoil.

If the second analysis shows that the topsoil has deteriorated in the nutritional requirements for topsoil the Contractor shall make good the topsoil by bringing it to the nutrient and organic status specified. Proposed additives shall be to the Architect's approval.

In the event that the quantity of existing topsoil does not meet with Contract requirements, the Contractor, on the instruction of the Architect, will make up the requirement using either imported C.D.G., soil conditioner, topsoil or topsoil mix, all as specified.

- (e) The Contractor shall ensure that topsoil heaps are properly maintained and that topsoil shall be placed in its final position within 24 months of importation to Site or, for site strip material, deposition for storage on Site. Weed control shall be carried out by spraying with approved weed killer.

The Contractor shall turn the topsoil heaps to prevent anaerobic conditions occurring.

The Contractor shall ensure that the topsoil which has been stored on Site meets the specification requirements in terms of nutrient and organic status at the time of planting. The Contractor shall arrange for tests to be carried out as specified in Clause 1.1(d).

If the topsoil stored on Site has deteriorated to such an extent that it is no longer possible to bring it to the specified levels by the addition of organic additives the Contractor shall make good by removing the unsuitable topsoil in tips off Site and replacing it with suitable topsoil material.

1.2 SOIL CONDITIONER, FERTILIZERS, SOIL BINDING AGENT AND MULCH

- (a) Soil Conditioner shall be manufactured product comprise with the following characteristics:
 - (i) Fine, free flowing consistency;
 - (ii) In stable condition;
 - (iii) Not capable of raising the temperature of the treated soil more than 5°C above the temperature of the untreated soil;

- (iv) Free from impurities and not made with materials that are known to contain pathogens or other toxic materials injurious to plants, humans, or animals;
- (v) Giving off neither toxic nor obnoxious fumes/odour;
- (vi) With a pH value between 6.5 and 7.5;
- (vii) With a moisture content when saturated between 30-50% (moisture weight as a proportion of overall weight);
- (viii) With an organic matter content of not less than 85% (dry weight);
- (ix) With a carbon nitrogen ratio between 20 and 30;
- (x) No weed growth, other foreign matter or contamination.

The Contractor shall present a certificate of analysis stating composition and physical and chemical characteristics. The analysis shall be carried out by a laboratory approved by the Architect or Architect's Representative.

A sample of 1 cubic meter shall be submitted to the Architect or Architect's Representative for approval before importation and use.

- (b) Organic compound fertilizers shall be stored in waterproof sealed bags under shelter, away from water and direct sunlight.
- (c) Pre-planting fertilizer shall be a granular, slow released, organic compound fertilizer with NPK analysis not less than 20% and organic matter not less than 30%.
- (d) Post-planting fertilizer shall be a granular organic compound fertilizer with NPK analysis not less than 30% and organic matter not less than 10%.
- (e) Hydroseeding fertilizer shall be a granular compound fertilizer with NPK analysis not less than 45%.
- (f) Soil binding agent shall be an approved binding medium applied in aqueous suspension by spraying onto the surface of the soil to stabilize and condition the soil. The binding agent shall in no way be injurious to plant growth.
- (g) Mulch shall be wood chips or tree bark. A sample of one cubic metre of mulch shall be submitted to the Architect or Architect's Representative for approval

before it is imported for use on site.

- (h) Peat shall be of partially decomposed fibrous or cellular stems and leaves of Sphagnum Mosses having a porous fibrous texture, fairly elastic and substantially homogenous with a pH value of not less than 4.5 and not greater than 6.0. It shall be baled and free of decomposed colloidal residue, wood, sulphur and iron, be brown in colour and finely shredded and suitable for horticultural purposes. Shredded particles shall not exceed 6mm in size.
- (i) Root activator shall be a chemical which contains plant hormones G.A., and IAA, and which can activate root growth.
- (j) Sealant shall be an approved fungicidal bituminous sealing compound or products having equivalent function or performance.

Water retaining crystals shall be a cross linked polyacrylamide copolymer. Or equivalent approved by the Architect.

1.3 SACK, BAGS, CONTAINERS ETC.

The Contractor shall retain for inspection by the Architect or Architect's Representative all sacks, bags, containers and the like in which fertilizer, mulch, grass-seed, pesticides, herbicides and the like are supplied and shall not dispose of these without the consent of the Architect or Architect's Representative.

1.4 FENCING

Protective fencing to existing vegetation shall be as specified in Clause 6.2.

1.5 TREES

- (a) A Light Standard Tree shall be a tree with all the following characteristics:
 - i. A stem diameter not less than 35mm when measured at a point one metre above the root collar.
 - ii. A strong, upright and reasonably straight unpruned stem well furnished with side branches.
 - iii. A branching head or a well defined straight and upright leader with branches growing out from the stem with good symmetry.
 - iv. A rootball not less than 350mm in diameter and 300mm in depth.
 - v. A total height above soil level between 1750mm - 2720mm.

(b) A Standard Tree shall be a tree with all the following characteristics:

- i. A sturdy straight stem not less than 1800mm in height between the soil level and the lowest branch.
- ii. A stem diameter not less than 60mm when measured at a point one metre above the root collar.
- iii. A well balanced branching head, or a well-defined straight and upright leader with branches growing out from the stem with good symmetry still providing a head.
- iv. A total height above soil level between 2750 - 3500mm.
- v. A rootball not less than 350mm in diameter and 300mm in depth.

(c) A Heavy Standard Tree shall be a tree with all the following characteristics:

- i. A sturdy, straight stem not less than 2100mm in height from soil level to the lowest branch.
- ii. A stem diameter not less than 100mm depending on species when measured at a point one metre from soil level.
- iii. A well balanced branching head, or a well-defined straight and upright leader with branches growing out from the stem with good symmetry still providing a head.
- iv. A total height above soil level greater than 3500mm.
- v. A rootball not less than 400mm in diameter and 400mm in depth.

(d) Containerized Standard, Heavy Standard and Semi-mature Trees and Palms, Heavy Palms Extra Heavy Palms and Feathered Trees shall have a container of dimensions not less than the rootball dimensions specified above.

II. REPARATORY WORKS

2.0 SETTING OUT

- (a) The Contractor shall be responsible for accurately setting out according to the Drawings all areas to be planted to the satisfaction of the Architect or Architect's Representative prior to the commencement of planting, and shall rectify errors in setting out at his own expense. Any discrepancy in Site area between that shown on the plans and the actual area on the ground shall be notified to the Architect or Architect's Representative as soon as it is discovered and prior to commencement of any relevant operations.

2.1 GROUND PREPARATION AND EROSION CONTROL

- (a) All areas to be planted shall be cleaned free of rubbish and weeds and all stones exceeding 50mm diameter. The Contractor shall ensure that all areas are free and kept free of any toxic substances.
- (b) Subsoil shall be formed by the Contractor to the required levels and contours with overall even compaction.
- (c) All obstructions to cultivation or deleterious material brought to the surface shall be removed from the Site by the Contractor, and any voids left by this operation shall be backfilled with subsoil as described. Ripping shall take place immediately before soiling operations in areas of topsoil spread. Any compaction resulting from delays in commencement of soiling shall be decompacted by the Contractor at his own expense.
- (d) In the case of slopes ripping shall take place parallel with the contours, not up and down the slope.
- (e) No topsoil shall be spread, or cultivation shall be carried out until the subsoil operations have been approved by the Architect or Architect's Representative. Topsoil shall be spread on the designated areas to the depth shown on the Drawings. The loose depth of the topsoil shall be sufficient to allow the area to conform to the levels shown on the Drawings after natural settlement has taken place whilst allowing for the 75mm of mulch which is to be spread to all shrub areas and any other areas indicated on the drawings. Placing and spreading of topsoil shall not take place during periods of heavy rain, nor when the topsoil is saturated. After settlement and compaction the finished level of topsoil

including the 50mm of mulch in the areas designated on the Drawings shall be 75mm below all kerb and footpath levels and all manhole inspection chambers, silt trap covers etc. After topsoiling, topsoiled areas are to be protected from further compaction and trafficking.

- (f) The Contractor shall take preventative measures to control erosion and siltation of all areas and shall restore or replace any portion of the Site which erodes, silts up or otherwise damaged.

2.2 CULTIVATION AND STONE PICKING

- (a) Cultivation shall take place only when the seeding or planting operation can begin immediately after cultivation, but no cultivation is to take place in weather or ground conditions where operations may destroy the soil structure.
- (b) Cultivation shall be carried out by approved mechanical or manual means to a minimum depth of 300mm.
- (c) After cultivation, stonepicking from the surface of soil areas shall be carried out whereby all stones over 50mm diameter are collected and removed off Site.

2.3 PROTECTION OF EXISTING VEGETATION

- (a) Where excavation is required adjacent to existing trees, precautions shall be taken in accordance with Clause 6.01.

2.4 TREATMENT OF EXISTING VEGETATION

WHERE INFILL PLANTING IS SPECIFIED

- (a) In areas of existing vegetation which is to be retained and augmented by the planting of trees, shrubs, whips and climbers, the existing vegetation shall be cut by hand (as Clause 2.4 (b)) to allow planting operations to be carried out.
- (b) Grass and other herbaceous plants shall be cut to a height of 50mm above ground level to prevent their competition with newly planted stock. Woody plants and small shrubs shall be cut to a height of 300mm above ground.
- (c) Shrubs and trees over 1500mm height shall be pruned according to Clause 6.00 where necessary, to allow planting operations to take place. The Contractor

shall obtain the approval of the Architect or Architect's Representative prior to any such works.

- (d) All rubbish and cut vegetation shall be removed off Site.
- (e) In areas of existing vegetation where footpaths are to be laid through the existing vegetation, care shall be taken in aligning the footpaths to avoid damage to all existing trees where possible. the Contractor shall peg out the alignment to the approval of the Architect or Architect's Representative before construction commences. Care shall be taken not to damage the roots of trees in the immediate vicinity of the footpath.

III. PLANTING WORKS GENERAL

3.0 WEATHER CONDITIONS FOR PLANTING

- (a) Planting shall take place in suitable weather conditions. Planting shall not take place in weather conditions which will result in initial drying out of root systems and / or scorching of leaves. Ideally planting shall take place in overcast or moist conditions. If planting has to be carried out in sun or drying winds, plants awaiting planting shall at all times be covered to prevent drying out in sun or drying winds.
- (b) The Contractor shall cease planting immediately when in the opinion of the Architect or Architect's Representative the weather conditions are not as defined above. Planting shall take place within the recognized planting season (1st February to 31st August), unless otherwise agreed in writing with the Architect.

3.1 ORIGIN OF PLANT MATERIALS

- (a) The Contractor shall state the origin of all plant materials, in good time before planting so that the Architect or Architect's Representative may inspect the nursery and agree on a selection of all plant material for approval. All plant material subsequently delivered to the Site shall be to at least the same standard or above in all respects as the approved. The Contractor shall note that in order to provide all the plant material as specified it may be necessary for him to grow the material in his own nursery in advance of commencing planting works on Site.

3.2 USE OF CHEMICALS

- (a) No chemicals shall be used on Site without the prior approval of the Architect or Architect's Representative. Such chemicals shall not be toxic to humans, birds or animals and shall be applied according to the manufacturer's instructions. The Contractor shall be entirely responsible for the storage, mixing, application and use of chemicals, including the proper disposal of product containers.

3.3 SUBSTITUTION OF PLANT MATERIAL

- (a) In the event of plant material as specified herein not being available, the Contractor shall notify the Architect or Architect's Representative immediately of available substitutes. No substitute shall be made without the prior written approval of the Architect or Architect's Representative.

3.4 MATERIALS TO BE AS SPECIFIED

- (a) All plant material shall be true to species and healthy, and shall not be less than

the minimum sizes specified. Plants having any habit or growth other than that specified shall be considered unacceptable.

3.5 PREPARATION OF PLANT MATERIAL FOR LIFTING, TRANSPORTATION AND STORAGE

- (a) Root pruning and undercutting of the root system of rootballed stock to the specified size of root-ball shall be carried out 12 months before lifting from the nursery. Plants grown in the open ground shall be well watered prior to lifting and shall be lifted carefully to ensure the specified root ball size is obtained. At the time of lifting, the root ball and the trunk from soil level to the lower branches of trees in the standard and heavy standard categories shall be securely wrapped to prevent loss of soil and moisture by using hessian or straw. The wrapping material shall not be removed until the plant is required for planting.
- (b) Containerized stock shall be well watered prior to despatch from the nursery and shall remain in the containers until required for planting.
- (c) All plants shall be wrapped and protected to prevent mechanical damage during lifting and transportation. This shall also be protected against excessive sunlight, wind and drought. Care shall be taken in packing to prevent over-heating with its resultant loss of foliage. Damaged plant material may be rejected by the Architect or Architect's Representative and the Contractor shall replace such damaged material. Damaged material which is not so rejected shall be carefully pruned using sharp clean implements to give a single flat sloping face cuts and wounds shall be painted with a fungicidal bituminous sealing compound, approved by the Architect or Architect's Representative.
- (d) Trees and shrubs which are not immediately planted in their permanent positions shall be stood upright on level ground, protected and maintained in good condition by the Contractor.

3.6 PLANTING

- (a) All plants shall be planted to accommodate the spreading root system of the plant to the same soil depth as in the nursery, and shall be well watered before removing them from containers. Plants are to be positioned upright and the soil firmed around the roots.

3.7 SUBSOIL TREE STABILIZATION SYSTEM

- (a) The method of tree stabilization system shall be subject to approval by the Architect or Architect's Representative before it is used.
- (b) All new tree planting shall be provided with tree stabilization system with 2-3 nos. wood deadmans below soil at planter bottom and tied to the root ball with stainless steel wire fixed on the deadmans and mounted on the root ball.
- (c) Tree ties shall be 7 strand stainless steel wire threaded
- (d) The catalogue and sample shall be submitted for approval

3.8 INSECT AND DISEASE CONTROL

- (a) The Contractor shall regularly check for any insect attack or fungal infestation particularly during known periods of activity. The Contractor shall report to the Architect or Architect's Representative any such occurrence and shall carry out remedial eradication by use of sprayed insecticide or fungicide, use in accordance with the manufacturer's instructions. Use of such sprays is to be with care and to have due regard to the safety and convenience of the general public. It shall be carefully controlled to avoid unnecessary dispersion, and shall be in accordance with Clause 3.02.

3.9 POSITIONING OF PLANTING MATERIAL

- (a) Tree and shrub areas shall be marked in outline with pegs spaced not more than 15m apart. The pegs shall be not less than 750mm long and 50mm in thickness and shall be firmly driven into the ground. The top 300mm of each peg shall be painted white.
- (b) The Contractor shall mark out the required planting interval with canes, stones chalk or other suitable markers along the longest edge of the area to be planted.
- (c) The first row of plants shall be the required distance from this edge and directly in line with each marker. In the case of planting areas edged by kerbs or walls, the first row of plants shall be planted as close to the edge as foundations will permit. In the case of planting areas adjacent to other planting areas, the first row of plants shall be planted at a distance which is half the specified planting distance for that species from the edge.

- (d) The second row shall be the required distance from the first. The pattern will be repeated over the whole planting area.
- (e) In the case of woodland mix and shrub mix planting, the plants shall be planted in positions in accordance with the matrix shown on the Drawings.
- (f) All species in whip mix and shrub mix planting are to be grouped so that between 9 and 19 plants of one species are planted in a distinct group (not in a straight line along a row).
- (g) The approximate numbers of plants to be planted per half day shall be set out by laying them down beside the hole in which they are to be planted. Plants shall not be removed from their containers until planting is taking place. All setting out shall be to the approval of the Architect or Architect's Representative.

3.10 NOTICES AND INSTRUCTIONS

- (a) In respect to Landscape Works, the Contractor shall give forty-eight hours notice to the Architect or Architect's Representative, of his intention to commence any one of the following operations : soiling, setting out, planting, hydroseeding, sprigging, turfing, fertilizing, visits to carry out Establishment Works.
- (b) The Contractor shall undertake any remedial landscape Works within twenty-four hours of notice by the Architect or Architect's Representative.

3.11 CULTURAL OPERATIONS AFTER PLANTING

- (a) During the period between planting and the notification by the Architect of the commencement of Establishment Works, the Contractor shall perform all works as specified for the healthy establishment of plants in accordance with the requirements of Clauses 7.00 to 7.01 inclusive.
- (b) The Contractor shall ensure that at the item of commencement of the Establishment Works, all plants and grass areas are in a healthy growing condition. The Contractor shall be responsible for any replacement planting which is necessary prior to the onset of the Establishment Period.

3.12 PROGRAMME FOR ESTABLISHMENT WORKS

- (a) The Contractor shall submit, prior to the commencement of Establishment Works, a programme indicating the frequency of visits and labor required to carry

out each of the operations defined as Establishment Works as specified in Clauses 7.00 to 7.09 inclusive.

3.13 LATIN NAMES

- (a) On planting plans, where Latin names and Chinese Characters are given for plants; the Latin name shall always take precedence.

3.14 SITE WORK SUPERVISION

- (a) The Contractor shall provide a full time supervisor during the implementation of site works. The duty of the supervisor shall monitor the soiling work, planting work and all planting-related work on site. The supervisor shall be a qualified arborist with a minimum 5 years related experience.

IV. PLANTING TECHNIQUES: TREES

4.0 NOTCH PLANTING OF TREES

- (a) Where Notch planting is required the notch shall be formed vertically by a hand held pick or spade, and shall be deep enough to accommodate the root of the plant. The plant shall be removed from the container and placed upright in the notch taking care to keep the soil and root intact. The plant shall be firmed in the ground with the shoot upright.

4.1 PIT PLANTING OF TREES

- (a) Where pit planting is required the pits shall be excavated to 100mm wider on every side than the root spread and to a depth of 100mm deeper than the root depth. Notwithstanding any other specified requirements, the minimum dimensions of planting pits shall be as given below unless otherwise agreed by the Architect.

Plant Type	Width and Length of Pit in mm	Depth of Pit in mm
Light Standard Tree	750	750
Standard Tree	750	750
Heavy Standard Tree	900	900

On slopes the depth will be measured from the lowest adjacent ground level on the downward side of the pit.

Any tree planted in a pit in a paved area shall have a minimum depth of 1200mm. The width and length of the pit shall be the maximum possible according to the dimensions of tree grille or tree grid.

- (b) Backfill material shall be mixed with 60g approved slow-release fertilizer per pit unless stated otherwise. In areas where topsoil or topsoil mix has already been spread back-fill material shall be the topsoil or topsoil mix which has been excavated from the pit mixed with soil conditioner as described. In cases where pits are deeper than the topsoil or topsoil mix already spread, the excavated subsoil shall be removed from site and the pit filled with topsoil mixed with conditioner as described.

Plant Type	Vol. of Soil Conditioner per pit (litres)
Light Standard Tree	30
Standard Tree	30
Heavy Standard Tree	50

- (c) In areas where pit planting is into hydroseeded areas with no topsoil or topsoil mix spread, back fill material shall be topsoil or topsoil mix mixed with soil conditioner as described in 4.1(b). Excavated material shall be removed from Site.
- (d) The pit shall be backfilled to a suitable level to maintain the same relationship between the root collar of the plant and the soil surface as existed in the nursery. The plant shall be set upright in the pit the roots spread carefully and backfilling with soil shall be completed, firming first with the knuckles.
- (e) In areas of turf or established grass before the pit is excavated, existing turf shall be cut from the position that the planting pit will take and replaced after the plant is in position, and the pit backfilled. If the turf is damaged during cutting and cannot be replaced whole, additional turf shall be brought onto the Site and used for this operation. Turf shall be pegged back in position at the top corners when laid on an incline.
- (f) In all backfilling operations adequate care shall be taken to avoid any damage to the root system, branches and leaves. The soil shall be consolidated in layers to ensure that the plant is firmly held in the ground and that air pockets are not left around the roots.
- (g) Individual planting pits for Light Standard and Standard Trees shall be excavated to 150mm wider on either side than the root ball width and 100mm deeper than the root ball depth. The bottom of the pit shall be broken up to a depth of 150mm. Material for backfilling purposes shall be topsoil or topsoil mix as specified. Backfill material shall be thoroughly mixed with 225g pre-planting fertilizer per tree. Excavated material shall be removed from site.
- (h) Individual planting pits for Heavy Standard Trees shall be excavated to 150mm wider on either side than the root depth and 75mm deeper than the root depth.

The bottom of the pit shall be broken up to a depth of 150mm. Excavated material is to be retained for backfilling the tree pit when necessary and when in the opinion of the Architect or Architect's Representative it is suitable. Material for backfilling purposes shall be topsoil or topsoil mix. Backfill material shall be thoroughly mixed with 225g of slow release fertilizer. Excavated material shall be removed from Site.

4.2 WATERING AFTER PLANTING

- (a) Immediately after planting, all plants shall be thoroughly watered with fresh water such that the roots of the plants are soaked.

1.3 MULCHING

- (a) Approved mulch shall be spread to a thickness of 75mm to all planting areas and any other areas as required on the Drawings dishing the mulch round the bases of the plants. The Contractor shall take care not to damage the plant material during mulching operations. Mulch shall be applied after planting and watering have taken place. Mulch shall be thoroughly dampened after spreading.

V. EXISTING TREES AND WOODLAND AREAS

6.0 PRUNING OF EXISTING TREES

- (a) Existing trees which are to be retained shall be pruned to remove dead or dangerous branches and to produce a well-shaped, balanced head. All work shall be carried out in accordance with good horticultural practice and B.S. 3998, 1996 “Recommendations for Tree Work” and shall be directed and supervised by the Architect or Architect’s representative. Safety precautions shall be taken to protect those engaged in operations as well as people and property in the vicinity. Pruning and removal of branches shall be done using sharp, clean implements to give a single flat, slopping face. Ragged edges of bark or wood are to be trimmed with a sharp knife. Large branches shall be removed in stages beginning with removal of the main weight of the branch with the final cut as close to the main stem as possible without damaging the bark. All cuts shall be made to avoid splintering or tearing of bark which would catch water and encourage rot. Twigs less than 15mm diameter may be cut with sharp secateurs.
- (b) Cracks, cavities or rotten wood shall be cut back to living tissue, sealed with fungicidal compound, and filled with an approved flexible filler to prevent ingress of water.
- (c) Any material pruning or treatment will be categorized as follows:
 - i. Hard prune
This shall include the removal of a substantial number of branches of up to 200mm in diameter in accordance with Clauses 6.0(a), 6.0(b) and 6.0(c) above.
 - ii. Light prune
This shall include the removal of a few branches up to 75mm in diameter in accordance with Clauses 6.0 (a), 6.0 (b) and 6.0 (c) above.
 - iii. Thin crown
This shall include the removal of overcrowded branches up to 50mm in diameter in the crown of the tree. Work carried out shall be in accordance with Clauses 6.0 (a), 6.0 (b) and 6.0 (c) above.
 - iv. Treat for pest and / or disease attack
This shall include the application of a suitable pesticide or fungicide approved by the Architect or Architect’s Representative to the infected areas in accordance with the manufacturer’s instructions.

6.1 PROTECTION OF EXISTING TREES

- (a) Where excavation is required near existing trees for construction of works, the following precautions shall be taken to protect the roots :
- i. roots exposed during excavation shall be wrapped with straw or hessian during construction of the works. Cutting of the roots shall be kept to a minimum.
 - ii. before backfilling, roots shall be cut cleanly back to living tissue and sealed with an approved bituminous sealant.
 - iii. excavation shall be backfilled with topsoil including sufficient slow release fertilizer to ensure a rate of application of 500g/cu.m.
- (b) Trench excavation for services, including drainage and sewage, shall be kept to a minimum of 1.5m from the tree trunk. Detailed location of services shall be agreed with the Architect or Architect's Representative before excavation commences if this minimum cannot be achieved. Large roots exposed in trench excavations and above the final line of the installation shall be preserved, and excavation close to trees shall be carried out with particular care to ensure this. Following installation of the services, damaged roots shall be cut back cleanly to living tissue and sealed with an approved bituminous sealant. Trenches shall be backfilled as specified, except that where topsoil is required, sufficient slow release fertilizer to assure a rate of application of 500g/cu.m. shall be applied.
- (c) Woodland areas shall be protected during the contract period by temporary bamboo fencing, as specified in Clause 6.2.
- (d) In respect of all existing trees and woodland the Contractor shall ensure, for the whole duration of the Contract, the following :
- iv. no unnecessary intrusion into areas of woodland or scrubland is made.
 - v. all access routes to construction areas which need to pass through woodland or scrub land shall be approved by the Architect's Representative.
 - vi. the limits of site clearance are to be agreed by the Architect or Architect's Representative on Site before site clearance commences. All trees to be cleared shall be marked by the Contractor and approved by the Architect or Architect's Representative before felling.
 - vii. no nails or other fixings shall be driven into trees.

- viii. no fencing or signs shall be attached to trees.
 - ix. no materials or machinery shall be stored under or against trees.
 - x. no workshop, canteens, or similar shall be installed beneath trees, nor shall equipment maintenance etc. be carried out under trees.
 - xi. no trees shall be used as anchors for ropes or chains used in guying, pulling and the like.
- (e) The Contractor shall exercise the greatest care during the progress of the work to avoid damage to any tree which the Contract does not require to be cleared.
- (f) As soon as the Site or any part thereof becomes available the Contractor shall erect Temporary Protective Fencing around each such tree or group of trees but no closer than 2m from the trunk of any such tree. The Contractor shall inform the Architect's Representative if Works are to be carried out within such fenced areas and, save with the express permission of the Architect or Architect's Representative or on his order, all such work shall be executed using only hand-held tools. The rates in the Contract shall include for this restriction.

6.2 TEMPORARY PROTECTIVE FENCING

- (a) The Contractor shall maintain the Temporary Protective Fencing in good repair and subsequently remove it. Removal will be subject to the permission of the Architect or Architect's Representative which will not normally be given earlier than the substantial completion of an adjacent and substantial part of the Works other than Landscape Softworks:
- (b) Provided that the Contractor may seek permission to remove the fencing temporarily if its removal is necessary for the satisfactory execution of the Works but he will be required to reinstate it as soon as possible.
- (c) Temporary Protective Fencing shall be 1200mm high comprising end straining posts with struts, intermediate posts with galvanized line wire, galvanized twisted wire and approved split bamboo poles.
- i. Staining posts shall be installed at corners, ends of runs and at intermediate positions 30 meters apart maximum. Posts shall be 1800mm long by 150mm diameter or 150mm square. Struts shall be 1700mm long by 150mm diameter or 150mm square and shall be housed and securely nailed to all straining posts in the direction of each

line of fencing.

- ii. Intermediate posts shall be installed at 2.5 meters apart maximum and shall be 1800mm long by 100mm diameter or 100mm square.
- iii. All posts and struts shall be of approved timber treated with approved preservative and shall be driven into the ground for a depth of 600mm minimum.
- iv. Two strands of 3.15mm high tensile galvanized steel wire to B.S. 4102 shall be strained and stapled with 38 x 4mm galvanized staples and fixed from 150mm from top of post and 150mm above ground level. Each line wire shall be strained tightly by means of a ratchet strainer. All line wires shall be secured to intermediate posts by one staple driven to a running fit and to straining posts by two complete turns round the post with the wire twisted back on itself and staples driven tightly into the post.
- v. Poles shall be hand-riven from bamboo poles approximately 30mm diameter. Pales shall be straight, pointed at the top as shown and notched 80mm from top and bottom.
- vi. The wire for wiring shall be not less than 2mm diameter galvanized mild steel wire conforming to B.S. 4102. Each line of wiring shall consist of two wires twisted together between the pales.
- vii. Poles to be positioned with not more than 50mm spacing between pales. One line of wire shall be fixed 80mm from the top of the pales and one 80mm from the bottom. Every 500mm top and bottom strained wires shall be bound to twisted wires on bamboo paling by means of wire twists of 2mm gauge galvanized wire.

VI. ESTABLISHMENT WORKS

7.0 ESTABLISHMENT WORKS - GENERAL

- (a) The Establishment period shall be 12 months starting from the date of Certification of Substantial Completion. During the period for Establishment Works, the Contractor shall carry out regular inspections and cultural operations as defined below to ensure that all trees thrive and were established. The Contractor shall keep the Site neat and tidy at all times. Unless otherwise specified (and / or exceptional weather conditions prevail), inspections shall be carried out at monthly intervals. The Contractor shall report to the Architect's Office before and after carrying out any Establishment Works and a record of the work carried out shall be submitted in duplicate forms provided by the Contractor and of a style approved by the Architect or Architect's Representative. The forms shall be countersigned by the Architect's Representative.

7.1 ESTABLISHMENT WORKS - WATERING

- (a) The Contractor shall water all trees as often as required to keep the ground moist all around the roots of the plants. An inspection of watering requirements shall be made in dry weather by the Contractor weekly. The Contractor shall thoroughly water areas as necessary to ensure the above conditions are achieved.
- (b) Fresh water only shall be used for the works. When required an analysis of water to be used shall be obtained by the Contractor for approval by the Architect or Architect's representative.
- (c) Water shall be applied by using an approved hose or sprinkler, drips, niggles and so as not to cause compaction or wash-outs of soil, or loosening of plants. The Contractor shall immediately make good any such damage.
- (d) The Contractor shall complete watering operations within 24 hours of an inspection which deems watering to be necessary.

7.2 ESTABLISHMENT WORKS - WEEDING AND LITTER REMOVAL

- (a) Planting in bare ground shall be maintained in a weed free condition by the removal of all unwanted vegetative growth over the whole planted area to the satisfaction of the Architect or Architect's Representative.
- (b) Planting not in bare ground shall be maintained by removing all competing and

overhanging weeds and by cutting all grass unwanted creepers as specified in Clause 7.04 and by keeping all areas within a 300mm radius of the base of each plant in a weed / grass free and tidy condition in accordance with Clause 7.02(a) above.

- (c) Weeding shall be carried out by a means approved by the Architect or Architect's Representative so as not to cause any damage to the Works. All weeds and rubbish resulting from this operation shall be removed off the Site. The Contractor shall weed areas as necessary and shall complete weeding within seven days of inspection.
- (d) All litter / rubbish in the planting areas shall be removed from the site. Litter / rubbish removal shall be completed within seven days of inspections.

7.3 ESTABLISHMENT WORKS - PRUNING

- (a) The Contractor shall prune all plants other than trees, when agreed with the Architect or Architect's Representative twice per year or more as required during the Establishment Period.
- (b) Pruning and removal of branches shall be carried out with sharp clean implements. Pruning shall be carried out from the cut just above, and sloping away to an outward facing healthy bud. Removal of branches shall be carried out by cutting flush with the adjoining stem and in such a way that no part of the stem is damaged or torn. Ragged edges of bark shall be trimmed with a sharp knife. Any cuts or wounds over 25mm diameter shall be painted with an approved sealant after trimming.

7.4 ESTABLISHMENT WORKS - GRASS CUTTING

- (a) The Contractor shall cut all grassed and not otherwise planted areas by approved mechanical or manual means so as to avoid root pulling. Grass shall be cut when it reaches a height of 100mm. Cutting shall reduce the height to 40mm.

7.5 ESTABLISHMENT WORKS - POST PLANTING FERTILIZER

- (a) The Contractor shall apply post-planting fertilizer when agreed with the Architect or Architect's Representative twice per year or more as required during the Establishment Period.
- (b) Each application of fertilizer shall be at the rate of 225g per Standard Tree and

Heavy standard tree.

- (c) Fertilizer shall be lightly worked into the soil surface around the base of the plant, allowing an even distribution. After application of fertilizer each plant shall be well watered.

7.6 ESTABLISHMENT WORK - FORKING OVER

- (a) The Contractor shall fork over the surface of all bare ground plants areas to relieve surface panning and compaction of the soil. The Contractor shall take care not to disturb the roots or loosen the plants. Any plants so disturbed shall be firmed up and well watered in immediately.

7.7 ESTABLISHMENT WORKS - SECURING STAKES AND TIES

- (a) The Contractor shall be responsible for securing stakes and ties if required or as requested by the Architect or Architect's Representative. An inspection shall be made every month by the Contractor and he shall replace all broken, damaged or otherwise unsatisfactory stakes and ties. Any ties which are causing chafing or abrasion of the trees shall be adjusted.
- (b) The Contractor shall be responsible for firming up any plants which become loose as a result of wind-rock or other cause.
- (c) The Contractor shall inspect the Site regularly for this purpose and after each storm or typhoon, to assess damage, which shall be reported to the Architect or Architect's Representative. Any damaged branches shall be carefully pruned and the wounds sealed.
- (d) The Contractor shall tighten or loosen the underground guys for the Semi-mature trees as required once towards the end of the Establishment period. The Contractor shall allow for uplifting and replacing the brick and granite paving blocks which are open jointed over the tree pit area. The Contractor shall also allow for topping up the tree pit with topsoil mix as described and bedding sand where necessary.

7.8 ESTABLISHMENT WORKS - REPLACEMENT OF PLANTS

- (a) The Contractor shall replace all plants which are dead, dying or otherwise unsatisfactory, if the cause is in the opinion of the Architect or Architect's Representative, as a consequence of the use of poor materials, workmanship, an

act of God or other such causes. Such replacement shall be to the relevant clauses of this Specification.

7.9 ESTABLISHMENT WORKS - MULCHING

- (a) The Contractor shall apply approved mulch when agreed and in the areas agreed by the Architect or Architect's Representative during the Establishment Period.
- (b) The Contractor shall allow for two applications of mulch each to a thickness of 75mm to be carried out during the Establishment Period. The final mulching operation is to be carried out in the last month of the Establishment Period.
- (c) Application of mulch shall be in accordance with Clause 4.3.

B.Living Green Wall

1.1 Scope of Works for Vertical Green Wall

1.2 The following vertical green wall type/system is to be supplied, designed and built by the Vertical Green Wall Sub-Contractor:

Code	System	Description	Location	Drawing No.
VG01 VG02 VG03	Vertical Green Wall (S.S. Panel System)	S.S. Panel with pre-grown climber plants	Building Facade	PP-1.01 PP-2.01 PP-3.01

2.0 Pre-grown climbers panels refer to climbers pre-grown on a reinforcement grid with rigid strength prepared off site for the installation of vertical greening systems, as specified in Drawings. Pre-grown Climbers Panels to exhibit all the following characteristics:

- Climbers to be grown off site at the Contractor's nursery in Hong Kong;
- Grown in a container not less than 200mm in diameter and 200mm deep; with panels of support/reinforcement which provides rigid strength, such as biaxial geogrids of polypropylene straps exhibit rigid strengths in each direction or equivalent. The aperture opening eye size should be minimum 30mm x 30mm and maximum of 50mm x 50mm. The Panels should be flexible enough for transport from Contractor's nursery in Hong Kong to the Site, as specified in Drawings in the form of roll for easy transportation;
- The size of each panel to be minimum 1,000mm wide and 2,500mm high;
- Climbers to be proposed should in minimum of five vigorous shoots not less than the specified height, species for each 1,000mm width panel should include choice of 2 from *Campsis grandiflora*, *Clerodendrum speciosum*, *Lonicera japonica*, *Parthenocissus himalayana*, *Podranea ricasoliana*, *Quisqualis indica*, *Thunbergia grandiflora*, *Wisteria sinensis* and additional plant species can be proposed and subject to the approval by Landscape Architect.
- A well-developed vigorous root system with root ball in the container to facilitate transportation to final location;
- The panels should be covered with at least 75% greenery by area.

1.3 General Responsibility of the Vertical Green Wall Contractor

The requirement for submission to the Buildings Department for approval are to be of Vertical Green Wall Sub-Contractor's full responsibility for compliance with all current regulations and codes based on the Landscape Architect's design intent drawings. No additional cost can be claimed due to increase in material size and/or thickness as per the requirement of Buildings Department, Government Authorities, Architect, Landscape Architect and Structural Consultant. A full set of shop drawings and product specification,

including design loadings, material properties and other information is needed for obtaining approval from the Buildings Department of Hong Kong, if required. No additional cost can be claimed due to the preparation and submission of shop drawings and product specification, if required.

2.1 Technical Requirement of Vertical Green Wall

2.2 The Vertical Green Wall Contractor shall submit the local job reference with project names, size and color photos of the completed green walls.

2.3 The Vertical Green Wall Contractor shall have practical experience and completed at least twenty successful projects, each green wall's size not less than 30m².

2.4 The Vertical Green Wall System should have at least 10 years of track record showing its stability.

2.5 The Vertical Green Wall System shall be of low maintenance and low water consumption.

2.6 The Vertical Green Wall System shall comprise of S.S. wire, growing medium, pre-grown plants, and supporting frame/structure.

2.7 The Vertical Green Wall System shall include all necessary structural calculations and drawings with RSE endorsement for government submission and approval, if required by the project team or Government Authorities. No additional cost can be claimed due to the preparation and submission of drawings / calculation.

2.8 Plants shall be pre-grown with the wire in nursery for at least 6 months before installation. The maintenance of the Pre-grown Climbers Panel at the Contractor's nursery in Hong Kong shall comprise the following:

- Establishment period for plant materials (from completion of installation of the panel at Contractor's nursery to end of the Contract, or as instructed by Landscape Architect):
During this period, the contractor shall, at its own cost, ensure that the vegetation is well maintained in accordance with Establishment Works.
- Maintenance period (from completion of installation of the panel at Contractor's nursery to end of the Contract, or as instructed by Landscape Architect):
The Contractor shall be responsible for the routine maintenance of the panel, including monitoring and inspection of the panel and replacement of defective component of the panel.

3.0 Submission Requirement

- 3.1** The Vertical Green Wall Sub-Contractor shall submit physical samples of the correct colour and finish together with the manufacturer's specification, technical and performance data for all materials and proprietary items for approval.
- 3.2** The Vertical Green Wall Sub-Contractor shall submit the complete manufacturer's technical literature for all proprietary products and materials including compliance with BS or other accepted standards.
- 3.3** Fabrication of the system shall commence only after formal approval is received from the Landscape Architect.
- 3.4** Irrespective of the materials and articles offered in the tender, all materials and articles for use of the works shall be in full compliance with the Specification unless otherwise agreed or endorsed by the Landscape Architect.
- 3.5** Plant photos submission with measurement tape to demonstrate the size and quality of plants.

4.0 Shop Drawings

- 4.1** The Vertical Green Wall Sub-Contractor shall submit shop drawings for all types of green walls within four weeks of contract commencement. Drawing submissions shall illustrate the contractor's detailed design including:
 - a) Plan, layout, details, levels, setting-out, detailed dimensions
 - b) Structural supports systems
 - c) Assembly and construction details
 - d) Plant pattern
 - e) Works by other trades and all other pertinent information
- 4.2** All shop drawing submissions shall be in both hard and soft copies.
- 4.3** Review of any data, approval of shop drawings shall not relieve the Sub-Contractor's responsibility for the performance of the works nor his responsibility to verify for himself that the work conforms to the Contract Documents.
- 4.4** Should the Vertical Green Wall Sub-Contractor consider it is necessary to fabricate or order any item before approval in order to meet completion date, it is entirely at his own risk and no compensation in time or cost will be granted if the material and/or workmanship are eventually not accepted by the Landscape Architect.

5.0 As-Build Drawings and Maintenance Manuals

- 5.1** Upon completion of the works, the Vertical Green Wall Sub-Contractor shall prepare a full set of as- built drawings and submit to the Landscape Architect for approval.
- 5.2** The Vertical Green Wall Sub-Contractor shall submit maintenance manuals for all types of green walls in both Chinese and English, with project name and location, including recommendations for periodic inspections, care and maintenance. Identify any common cause of damages, with instructions for temporary patching, until permanent repair can be made.

6.0 Establishment Works

- 6.1** The Vertical Green Wall Sub-Contractor shall provide 12-months Establishment Period / Defects Liability Period for all the Green Wall Systems starting from the Date of Certificate of Substantial Completion. The Vertical Green Wall Sub-Contractor shall carry out regular inspections to ensure that all plants thrive and established. The Vertical Green Wall Sub-Contractor shall keep the site in a neat / tidy and free from litter and rubbish at all times. Establishment works shall include but not limited to the followings:
- a) Regular trimming of plants
 - b) Clean and tidy up all plants
 - c) Replacement of unhealthy / damaged / dead plants
 - d) Application of fertilizer
 - e) Checking / repairing the irrigation system
 - f) Monthly site inspection
 - g) Submission of Monthly Maintenance Report. The report shall address all works and operations performed and completed during the reporting month. Record photographs for the full extent of each green wall shall be provided in the report. The report should be submitted in hard copy and the record photographs shall be colored date-stamped. Format of the report to be agreed by the Landscape Architect. All reports should be submitted within 14 calendar days after each calendar month.
 - h) Submission of incident report after each inclement weather or as instructed.

7.0 Tender Submission

- 7.1** The tenderer shall submit the followings during tendering stage:
- a) Company Organization Chart
 - b) Project Organization Chart and CV of the Key Personnel

- c) Proposal of the green wall system which shall include all necessary components, such as modular tray/box, growing medium, irrigation system, drainage system etc.
- d) Method Statement for installation of green wall
- e) Preliminary Program of Works, which should make reference to OP date and CC date
- f) Preliminary Maintenance Program and Schedule
- g) List of previous Job reference
 - Project Name / Location / Site Area
 - Client / Contract Sum / Year of Completion
 - Color Photographs
- h) Plant photos for all the specified plants in the Contract (if any)